S1.1 Introduction to the particulate nature of matter

Elements, compounds and mixtures

Learning outcomes

By the end of this section you should be able to distinguish between the properties of elements and mixtures.

Almost everything we know about atoms has been discovered in the past 100 years or so, but it was Democritus who in 400 BCE in ancient Greece came up with the first known model of an atom. He theorised that matter could be cut up into smaller and smaller pieces until a point was reached where it could not be cut up anymore. He called these tiny pieces *atomos*, which means indestructible in Greek. It was not until the 1950s, however, until atoms were seen for the first time through an electron microscope (**Figure 1**).

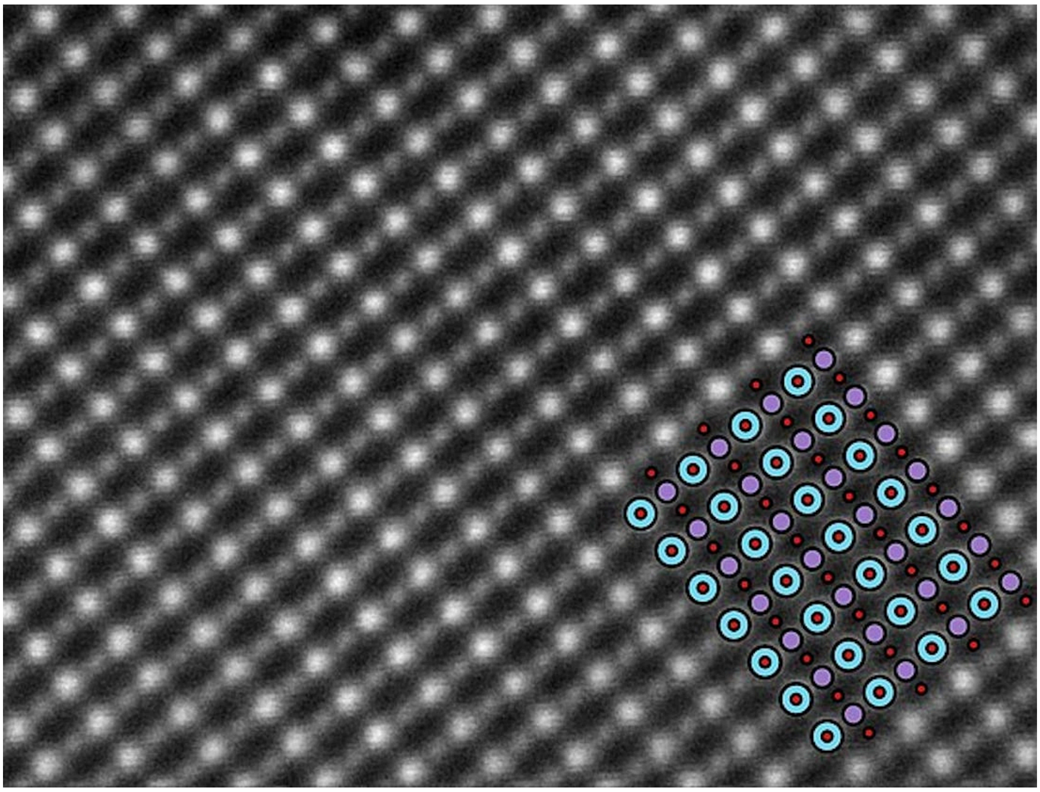

Figure 1. An image of atoms seen under an electron microscope.

How does Democritus' model compare to the actual image of an atom? I am sure you will agree that it looks quite similar to the description given by Democritus some 2500 years ago, although we now know that atoms are not indestructible and are actually made up of smaller particles called sub-atomic particles. Based on this, how accurate are the representations of particles and atoms that are used in this subtopic? Does one model explain all? Do they help with your understanding or can they introduce misconceptions? Keep these questions in mind as you read on.

Nature of Science

Aspect: Theories

An important point about theories is that they change over time. The concept that all matter is made up of very small particles, or atoms, is many thousands of years old. However, the ideas involved in 'atomism' remained undeveloped and unaccepted for long periods of time.

In 1803, the British chemist John Dalton developed an atomic theory to explain why chemical elements reacted in simple proportions by mass. He proposed that each chemical element consisted of identical atoms and that these atoms could bond together to form chemical compounds. Dalton's interest in atoms was strongly influenced by the experiments he performed on gases, and it is not clear whether he was aware of these previous ancient ideas. He didn't prove the existence of atoms but simply demonstrated atomic theory was consistent with experimental data.

In 1897, the British physicist J. J. Thomson discovered the electron. Based on his studies, he proposed a model of the atom, known as the 'plum pudding' model, in which the electrons were embedded in a sphere of positive charge.

In 1904, the Japanese physicist Hantaro Nagaoka rejected Thomson's theory on the grounds that opposite charges are impenetrable — they cannot exist within each other. Nagaoka proposed an alternative model of the atom in which a positively charged centre is surrounded by a number of orbiting electrons, similar to Saturn and its rings. This idea was eventually rejected, even by Nagaoka himself, but it showed how the idea of the atom having a massive nucleus and orbiting electrons was developing amongst scientists.

In 1909, the New Zealand physicist Ernest Rutherford demonstrated that the atom is, in fact, largely empty space. This is covered in more detail in section S1.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/).

In 1914, the model of the atom was further developed by Danish physicist Niels Bohr. Bohr proposed that electrons can only exist in certain fixed energy levels and that they can transition between these energy levels by absorbing or emitting exact amounts of energy. This concept is explored in more detail in section S1.3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/).

We can classify matter into two broad categories: pure substances and mixtures.

Elements and compounds are pure substances meaning that they are made up of one type of substance and have a fixed composition. Mixtures are made by combining two or more pure substances together. Therefore, they do not have a fixed composition and are not classified as pure substances. In this section, we will look at each of these in more detail, starting with elements.

Elements

Elements are often described as the building blocks of matter. They are the simplest forms of matter and consist of only one type of atom. For this reason, elements cannot be broken down chemically into simpler substances. Iron (symbol Fe), for example, is an element that is made up of only iron atoms (**Figure 2**). Note that the atoms in the iron are arranged in fixed positions which is characteristic of a solid. The anvil in **Interactive 1** is made of iron and the animation shows the atoms in the solid metal slowly vibrating in their fixed positions.

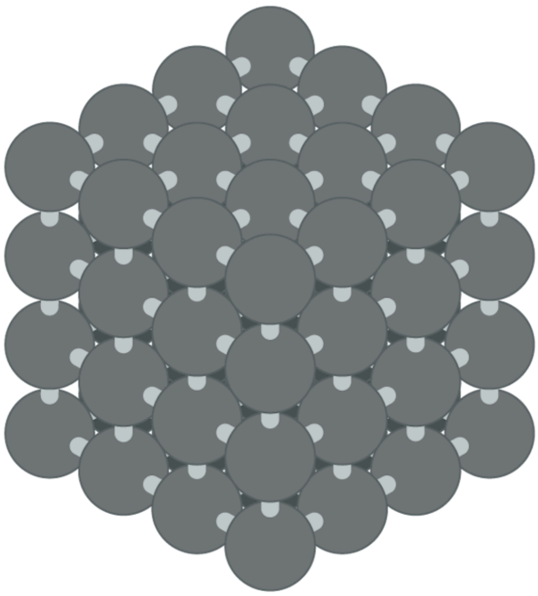

Figure 2. The arrangement of atoms in solid iron.

More information for figure 2

The image shows a drawing illustrating the atomic structure of iron. The atoms are depicted as grey spheres, arranged in a regular cubic pattern, representing a characteristic crystal structure. Each sphere connects horizontally and vertically to the adjacent spheres, forming a uniform, repeating lattice. This arrangement signifies the fixed positions of atoms typical in a solid state.

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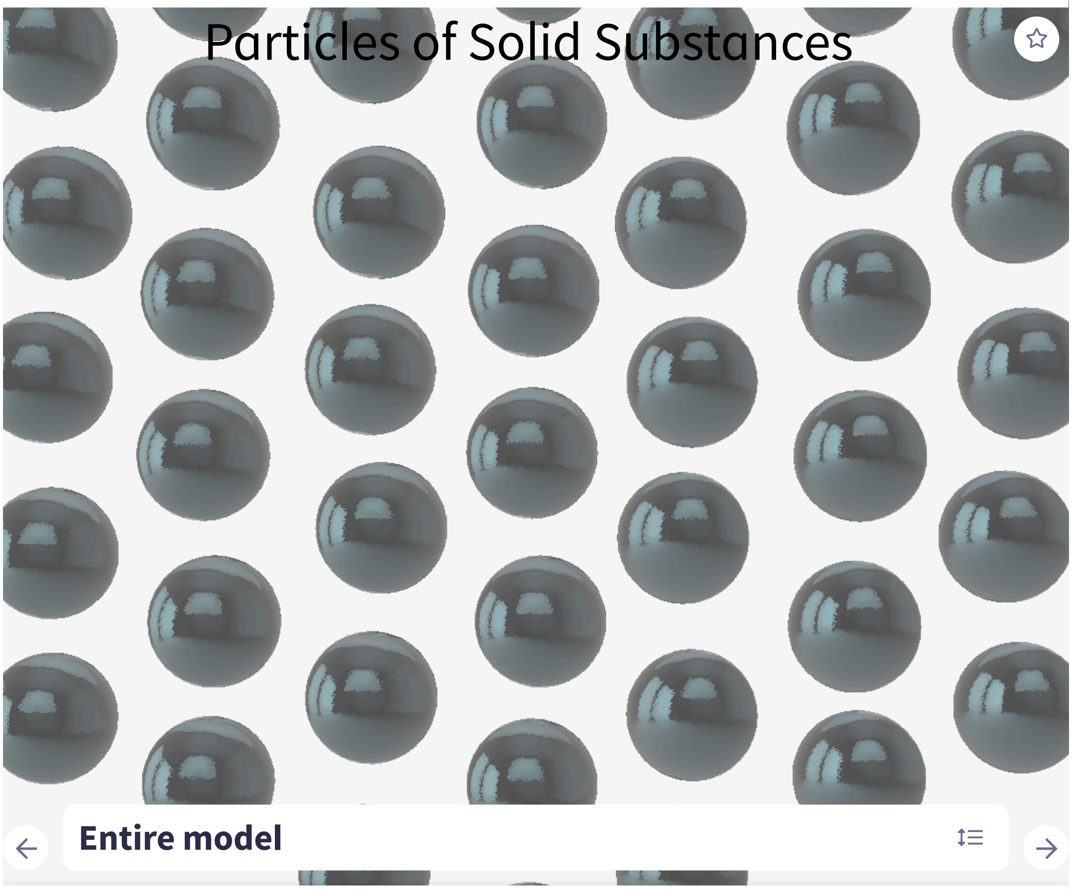

Introduction Share Capture Augmented Reality Language

Interactive 1. A close-up view of the atoms in an anvil made of iron.

More information for interactive 1

This interactive animation model helps users understand the behavior of particles in a solid substance. The model showcases an anvil made of iron, illustrating how its particles are arranged in a hexagonal pattern. As users interact with the model, they can observe the particles slowly vibrating in place, which demonstrates how solid particles are tightly bound and only able to oscillate within a fixed range.

The interactive features allow users to zoom in on the anvil by selecting the play option, offering a closer look at the arrangement and movement of the particles. A seek bar with a play/pause button at the bottom lets users control the animation, and a toggle button for "repeat" and "skip forward" is available for easier navigation.

In addition, a carousel selector at the bottom lets users explore different parts of the model, and the "Introduction" option provides a brief explanation: "Substances in a solid state consist of particles that are tightly bound in fixed positions. These particles can only oscillate or vibrate within a limited range around their fixed points, which gives solids their defined shape and structure."

This interactive model helps users gain a deeper understanding of the microscopic behavior of particles in solids, offering a visual way to grasp fundamental concepts in material science and physics.

A list of all known elements can be found on the periodic table. There are 92 naturally occurring elements with the remainder being synthesised since the mid-1940s. New elements are still being synthesised with the most recent one being Nihonium (symbol Nh) which was synthesised in the early 2000s.

Elements can exist as individual atoms, or they can exist as atoms of the same element bonded together. For example, hydrogen, H_2, exists as two hydrogen atoms bonded together to form a molecule. These are still classified as elements because they consist of only one type of atom.

Study skills

The following examples are elements that are always found in nature to exist bonded together as molecules, rather than as individual atoms. These can be diatomic elements (formed by two atoms of the same element bonded together) or polyatomic elements (more than two atoms of the same element bonded together).

Examples of the diatomic elements are: hydrogen (H_2), nitrogen (N_2), oxygen (O_2), fluorine (F_2), chlorine (Cl_2), bromine (Br_2) and iodine (I_2).

Examples of polyatomic elements are: phosphorus (P_4) and sulfur (S_8).

Elements such as carbon can also exist as allotropes which are different forms of an element in the same physical state. The allotropes of carbon will be covered in more detail in section S2.2.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/).

Compounds

Compounds are pure substances composed of two or more different elements chemically combined in fixed ratios. Carbon (symbol C) is an element that forms millions of different compounds and even has its own branch of chemistry dedicated to it, known as organic chemistry. One such compound that contains carbon is methane which has the molecular formula CH_4. Methane is made of two different elements, carbon and hydrogen (symbol H), chemically bonded in fixed ratios (**Figure 3**). For every one carbon atom in a molecule of methane, there are four hydrogen atoms.

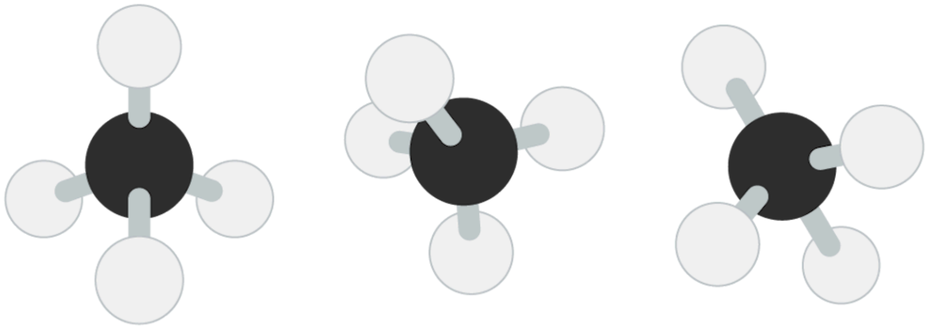

Figure 3. The ball and stick model for methane, CH_4.

More information for figure 3

The image shows a ball and stick model of methane (CH4), illustrating its molecular structure. It consists of three identical models, each with a central black sphere representing a carbon atom. Four white spheres are connected to each black sphere by thin rods, representing hydrogen atoms bonded to the carbon. The models are depicted in a three-dimensional arrangement to show the tetrahedral geometry of methane. Each hydrogen atom is equidistant and symmetrically placed around the carbon atom, highlighting the balanced spatial configuration typical of methane molecules.

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Another example of a compound is water, which has the molecular formula H_2O. Water is composed of hydrogen and oxygen atoms in a 2:1 ratio (**Figure 4**).

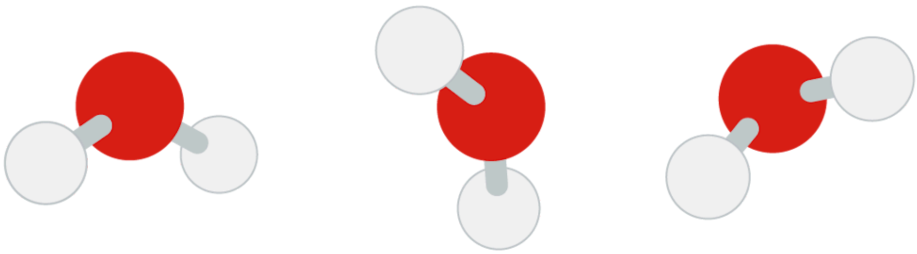

Figure 4. The ball and stick model for water, H_2O.

More information for figure 4

The image depicts a ball-and-stick molecular model of water (H2O). In this model, each water molecule is illustrated with a central red sphere representing an oxygen atom. Two smaller white spheres connected to the red sphere represent hydrogen atoms. The red sphere is larger, visually indicating the greater size or mass of the oxygen atom compared to the hydrogen atoms. The connections between the red and white spheres signify the chemical bonds, highlighting the molecular structure of water in a 2:1 hydrogen to oxygen ratio. This type of diagram is commonly used to demonstrate molecular geometry and the interactions between atoms in a compound.

[Generated by AI]

The atoms in a compound are chemically bonded to one another so they cannot be separated using physical methods. The properties of compounds are quite different from the elements they contain.

This point is illustrated in the reaction between the elements sodium and chlorine to produce the compound sodium chloride (NaCl) (**Figure 5**). Sodium is a very reactive metal that reacts vigorously with water. Chlorine is a toxic gas that is used to kill bacteria in water treatment. The compound sodium chloride is a relatively safe substance that has been used for centuries to flavour food. The properties of sodium chloride are very different from the elements that it is made from.

Making connections

In the DP chemistry course, you will learn about three types of bonding: ionic, covalent and metallic. These different types of bonding result in different structures and properties. These will be covered in more detail in subtopics S2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/), S2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/) and S2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43111/).

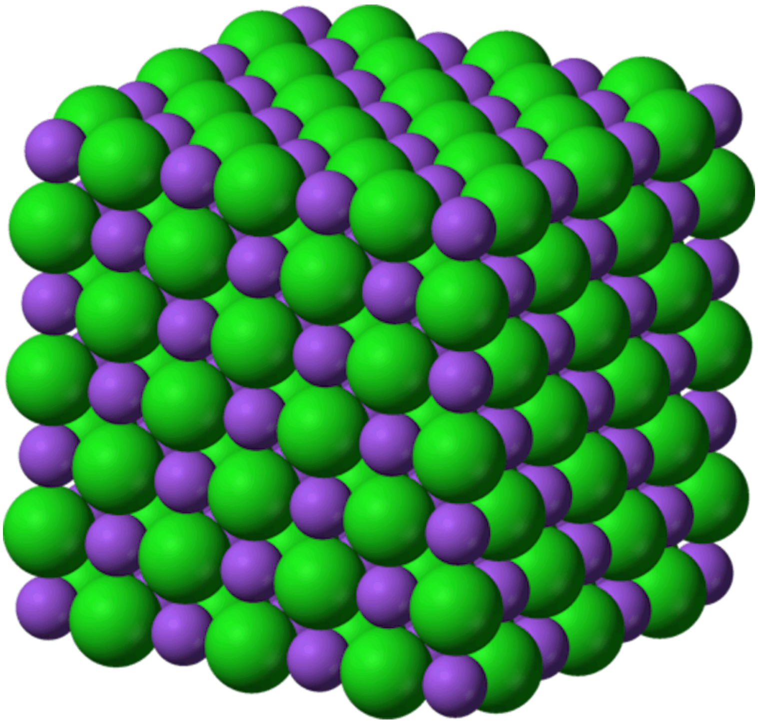

Figure 5. The lattice structure of sodium chloride, NaCl.

Source: "Sodium-chloride-3D-ionic (https://commons.wikimedia.org/wiki/File:Sodium-chloride-3D-ionic.png)" by Benjah-bmm27 is in the public domain

More information for figure 5

The image depicts an atomic structure model of sodium chloride (NaCl), commonly known as table salt. The model shows a three-dimensional arrangement of atoms in a cubic lattice. Purple spheres represent sodium atoms, while larger green spheres represent chlorine atoms. These atoms are arranged in a regular pattern, forming a cube with alternating purple and green spheres both horizontally and vertically. This structure demonstrates the ionic bonding pattern typical of sodium chloride, where each sodium ion is surrounded by six chloride ions and vice versa. The regular alternation and cubic symmetry illustrate the repeated pattern of ions in the crystal lattice, which is essential for understanding the compound's physical properties.

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You may have noticed that the compounds illustrated in this section have different structures. The structure of a compound depends on the type of bonding between its atoms. These bonding types will be covered in more detail in subtopic S2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/).

Drag and drop each of these substances into the correct column in **Interactive 2**.

Element	Compound

S	C_2H_6	CO_2	Mg	Cl_2
Li	LiCl	H_2	NH_3	CH_3OH

Check

Interactive 2. Element or compound?

Mixtures

The final category of matter is mixtures which, unlike elements and compounds, are not classified as pure substances.

- Mixtures are composed of two or more elements or compounds in no fixed ratio.
- Mixtures contain pure substances that are not chemically bonded so they can be separated by physical methods. These physical methods will be covered in the next section.
- The components of a mixture retain their individual properties. An example of this is a mixture of sugar and water or iron and sulfur (**Figure 6**). The iron retains its magnetic properties when

mixed with the sulfur, and this property can be used to separate the mixture.

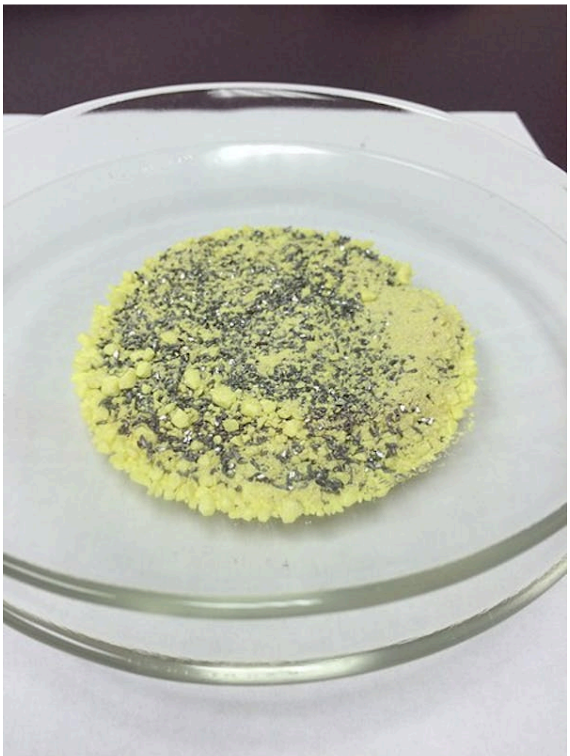


Figure 6. A mixture of sulfur (yellow) and iron filings.

Making connections

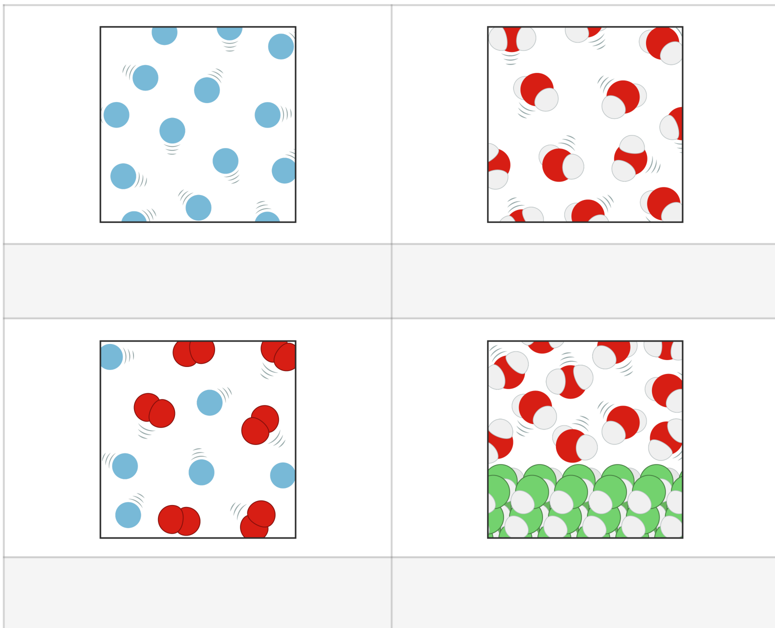
Alloys are considered to be mixtures even though the components of the mixture are bonded by metallic bonds. Alloys and metallic bonding are covered in more detail in subtopic 2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43111/).

Mixtures can be further categorised as being either homogeneous or heterogeneous. A mixture of oil and water is an example of a heterogeneous mixture (**Figure 7**). This type of mixture has visible phases or boundaries such as those that can be seen between the oil and the water. The composition of a heterogeneous mixture is non-uniform, meaning that different parts of the mixture have a different composition. Other examples of heterogeneous mixtures include carbonated water, which has bubbles of gaseous carbon dioxide mixed with water or the mixture of iron and sulfur discussed previously in this section. The iron and sulfur are clearly visible in the mixture as two visible phases.

Figure 7. A heterogeneous mixture of oil and water (left) and a homogeneous mixture of salt and water (right).

Homogeneous mixtures do not have visible phases or boundaries. They have a uniform composition meaning that the components of the mixture are equally distributed and in the same state, making the composition the same throughout the mixture. An example of a homogeneous mixture is the air that we breathe. The components of air such as gaseous nitrogen, oxygen and argon are evenly mixed resulting in a uniform composition. Another example of a homogeneous mixture is salt water. The salt is known as the solute and the water as the solvent. When the salt is dissolved in the water, it is known as an aqueous solution, or simply a solution. Salt dissolves in water because it is soluble in water. Solvation is the process where the solvent particles surround and interact with the particles of the solute. Substances that do not dissolve are known as being insoluble. The concept of solubility will be covered in more detail in section S1.4.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/).

Check your understanding by dragging each term to the correct image in **Interactive 3**.



Heterogeneous mixture

Element

Homogeneous mixture

Compound

Check

Interactive 3. What am I?

Creativity, activity, service

Strand: Service
Learning outcomes:

- Demonstrate how to initiate and plan a CAS experience
- Demonstrate engagement with issues of global significance

The world's seas and oceans can be thought of as either homogeneous or heterogeneous mixtures. If we consider the seawater alone, it is a homogeneous mixture; however, if we consider the living organisms that also live in the aquatic environment, then it is a heterogeneous mixture.

There is also a large amount of waste dumped into the seas and oceans. It is estimated (https://www.unep.org/interactives/beat-plastic-pollution/?gclid=Cj0KCQiA5NSdBhDfARIsALzs2EDrn-bzZc7H6pZNximZhy4FDvSba1R_pV-XFNEwPCRV2wHUoQZValgaAlz-EALw_wcB) that there are between 75 000 000 and 200 000 000 tonnes of plastic in the oceans. This plastic has a detrimental effect on wildlife as well as the possibility of ending up in the human food chain (https://www.foodstandards.gov.au/consumer/generalissues/Pages/Microplastics-in-food-.aspx). A potential CAS experience could be to set up and run a recycling programme at your school or in your local community to reduce the amount of plastic waste that ends up in the world's oceans.

Try the activity below to check your learning on elements, compounds and mixtures.

Activity

- **IB learner profile attribute**: Knowledgeable
- **Approaches to learning:**
  - Thinking skills — Reflecting on the credibility of results.
  - Social skills — Actively seeking and considering the perspective of others
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/group activity

1. Complete the interactive table by clicking on the any cell you believe to be true. Some rows have more than one correct answer.

Interactive 4. Do you know about substances?

More information for interactive 4

An interactive table with four checkboxes for each condition allows users to mark whether it is correct or incorrect for Elements, Compounds, Heterogeneous mixtures, and Homogeneous mixtures.
The conditions listed on the left are as follows:

1. Are pure substances
2. Composed of two or more different elements chemically bonded in fixed ratios
3. Composed of two or more different elements or compounds, not in fixed ratios
4. Components are not chemically bonded.
5. Can be separated by physical means
6. The components retain their individual properties
7. Have a uniform composition throughout
8. Have visible phase boundaries
9. Have exact melting or boiling points

Users can select the appropriate checkboxes, which indicate whether the selection is correct or incorrect. When you select the Reset button on the top, an Error message appears. The full-screen icon is at the bottom right.
Read below for the solution:

1. Elements and compounds are pure substances.
2. Compounds are composed of two or more different elements chemically bonded in fixed ratios.
3. Heterogenous mixtures and homogeneous mixtures are composed of two or more different elements or compounds, not in fixed ratios.
4. The components of heterogenous mixtures and homogeneous mixtures are not chemically bonded.
5. Heterogenous mixtures and homogeneous mixtures can be separated by physical means.
6. The components of heterogeneous mixtures and homogeneous mixtures retain their individual properties.
7. Elements, compounds, and homogeneous mixtures have a uniform composition throughout.
8. Heterogeneous mixtures have visible phase boundaries.
9. Elements and compounds have exact melting or boiling points.

This interactivity allows users to explore the properties of elements, compounds, heterogeneous mixtures, and homogeneous mixtures, highlighting their similarities and differences.

2. For each of your choices above, explain your reasoning to a partner.

The concept diagram in **Figure 8** below summarises the organisation of matter into pure substances and mixtures.

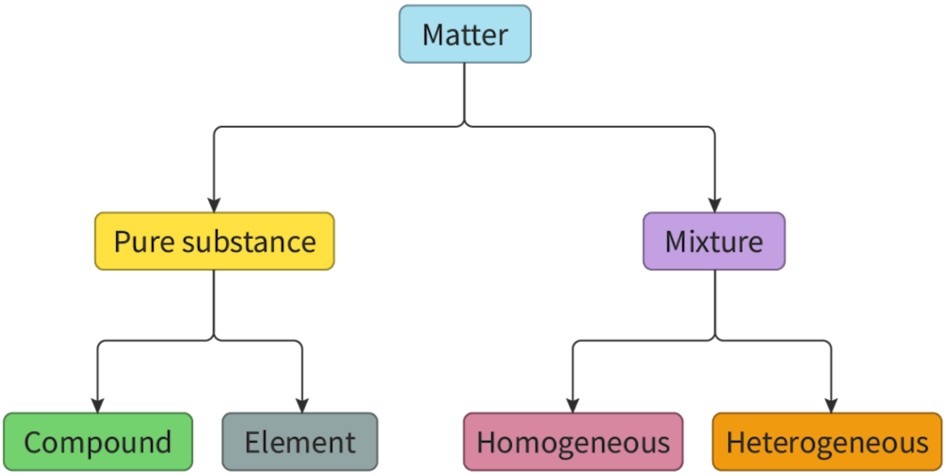


Figure 8. Matter can be divided into pure substances or mixtures.

More information for figure 8

The diagram is a branching flowchart illustrating the classification of matter. At the top level, there is a single box labeled "Matter." This box branches into two boxes at the second level labeled "Pure Substance" on the left and "Mixture" on the right. These boxes further branch at the third level. Under "Pure Substance," there are two boxes labeled "Compound" on the left and "Element" on the right. Under "Mixture," there are two boxes labeled "Homogeneous" on the left and "Heterogeneous" on the right. Arrows indicate the hierarchical relationship from top to bottom, illustrating how matter is divided into either pure substances or mixtures and further subcategories.

[Generated by AI]

S1.1 Introduction to the particulate nature of matter

Separation techniques

Learning outcomes

By the end of this section you should be able to distinguish between the properties of elements and mixtures.

The Earth is sometimes known as the blue planet because of its abundance of water. Water is essential to life but according to the Centers for Disease Control and Prevention (http://www.cdc.gov/healthywater/global/wash_statistics.html#:~:text=Find%20information%20on%20how%20safe,h (CDC), approximately two billion people do not have access to clean drinking water at home. How can this be the case when nearly two-thirds of the planet is covered in water? The answer lies in the composition of the water in the seas and oceans. Rather than being drinkable water, it is a mixture of salt and water with approximately 35 grams of dissolved salt per litre. This amount of salt is much higher than the human body can tolerate and drinking sea water would quickly prove fatal. To produce drinkable water from sea water requires the separation of the salt and water. One such technique is evaporation, which will be covered in more detail later in this section.

One challenge for the future is to provide clean and accessible drinking water for the Earth's increasing population which is one of the Sustainable Development Goals (https://sdgs.un.org/goals) (SDGs) proposed by the United Nations (UN). As an IB student, can you think of any ways in which you can help meet this goal?

In the previous section we learned that because the components of a mixture are not chemically bonded, they can be separated relatively easily using physical methods. In this section we will look at different separation techniques that can be used to separate the components of homogeneous and heterogeneous mixtures into pure substances.

Practical skills

Tool 1: Experimental techniques — Applying techniques

What factors are considered in choosing a method to separate the components of a mixture? Does the type of mixture, or the components of the mixture, determine the separation technique used?

Filtration

Filtration involves the separation of an insoluble solid from a liquid or solution. An example would be the separation of a mixture of sand and water, which is a heterogeneous mixture (**Figure 1**). The mixture of sand and water is poured through a piece of filter paper held in a filter funnel. The sand is not dissolved in the water and cannot pass through the pores in the filter paper. The water, known as the filtrate, is able to pass through the pores and is collected in the beaker. The sand, known as the residue, remains in the filter paper and can be collected.

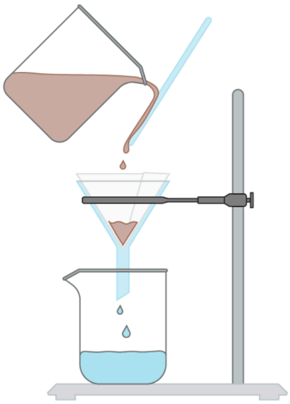


Figure 1. Filtration of a mixture of a liquid and an insoluble solid.

More information for figure 1

The diagram illustrates a laboratory setup for the filtration process. At the top, there is a beaker with brown liquid being poured onto a glass rod that directs the liquid into a funnel. The funnel contains a piece of filter paper, positioned to collect the mixture. As the liquid enters the funnel, brown residue collects at the bottom of the filter paper, while clear liquid passes through and drips into a receiving beaker below. The setup is held by a stand, supporting both the funnel and the receiving beaker. This visual demonstrates the separation of a mixture where the solid remains as residue and the liquid becomes the filtrate.

[Generated by AI]

Evaporation

Evaporation is used to separate a mixture which has a solute dissolved in a solvent which is usually water. The solution is heated in an evaporating dish and the solvent evaporates leaving the solute behind. An example of this is salt water, a homogeneous mixture. The mixture of salt and water is heated, and the water evaporates, leaving the salt behind **(Figure 2)**.

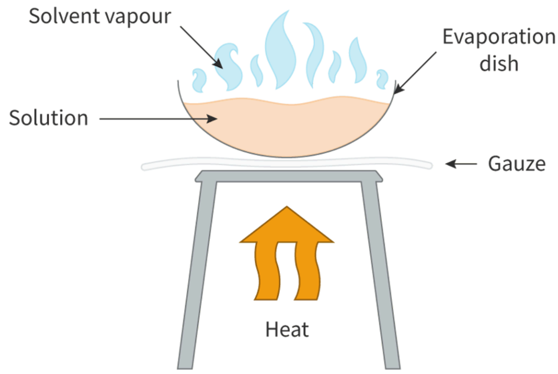


Figure 2. The separation of an aqueous solution using evaporation.

More information for figure 2

The diagram illustrates the process of evaporation. An evaporating dish containing a light orange solution is placed on a gauze-covered tripod. Heat is applied from below, indicated by an upward orange arrow. Blue wispy arrows are shown rising from the dish, representing solvent vapor. The diagram is labeled with 'Solvent vapour,' 'Solution,' 'Evaporation dish,' and 'Heat,' pointing to the respective elements. This setup demonstrates how a solution, such as saltwater, can be heated to evaporate the solvent, leaving the solute behind.

[Generated by AI]

Solvation

Solvation involves the separation of a heterogeneous mixture of two solids based on differences in solubility if one of the substances is soluble in a solvent, but the other solid is insoluble. During solvation, the solvent molecules (most often water) surround the soluble molecules and dissolve the solid into a solution. The insoluble solid can now be separated by filtration. The soluble substance can be separated from the solution by evaporation.

Distillation

Distillation involves the separation of a liquid mixture based on the difference in volatility or boiling points between the components of the mixture. An example is the separation of a mixture of ethanol and water. Water and ethanol are miscible, meaning that they fully dissolve when mixed together, forming a homogeneous mixture. Each liquid has a different boiling point; water boils at 100 °C and ethanol at 78 °C. When a mixture of water and ethanol is heated, the ethanol will evaporate first because of its lower boiling point and it will rise up the distillation column. It then passes through the condenser, which allows the ethanol to cool and condense back to a liquid to be collected in a flask (**Figure 3**).

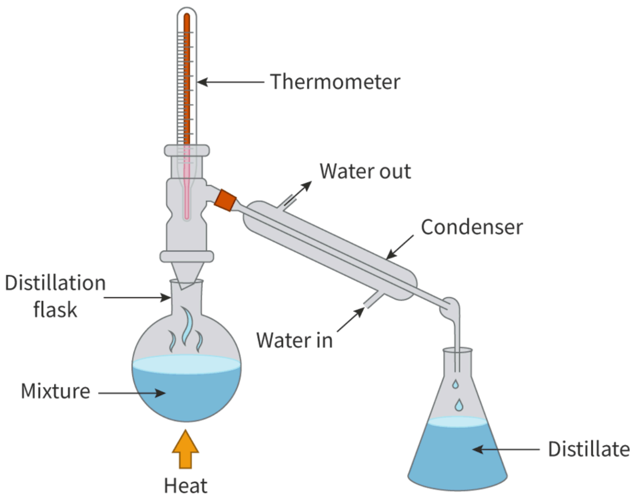


Figure 3. Apparatus for carrying out distillation.

More information for figure 3

The image is a diagram illustrating a distillation apparatus setup. At the base, there is a round-bottom distillation flask containing a liquid mixture, indicated by a wavy line representing heat being applied to the bottom. Above the flask, a thermometer is inserted into an adapter to measure temperature changes.

To the right, a condenser is connected to the side arm of the adapter. The condenser consists of a small inner tube and a larger outer tube through which water circulates. There are labels showing 'Water in' and 'Water out' to indicate the flow direction.

From the end of the condenser, droplets of blue liquid, identified as 'Distillate,' drip into a receiving conical flask positioned on the right. Each component of the apparatus is clearly labeled, including the distillation flask, condenser, and the distillate.

[Generated by AI]

Distillation is used to separate the components of crude oil into gasoline, kerosene, and lubricating oil (**Figure 4**).

Figure 4. Fractional distillation is used to separate the components of crude oil at an oil refinery.

Chromatography

Chromatography is used to separate the components of a mixture. The four types of chromatography are paper chromatography, thin-layer chromatography, gas chromatography and high-performance liquid chromatography. In the DP chemistry course, students are only required to know about the first two types - paper chromatography and thin-layer chromatography (TLC). In this section, we will focus mainly on paper chromatography as it is the easiest type of chromatography to do in the laboratory. Both types of chromatography are covered in more detail in section s2.2.10 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chromatography-id-45133/) and section 1.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/paper-or-thin-layer-chromatography-id-48918/).

Paper chromatography uses a piece of chromatography paper and a solvent. The paper is known as the stationary phase and the solvent, the mobile phase. The components of the mixture move through the stationary phase at different rates due to differences in solvation. Those components with a greater affinity for the mobile phase have a greater interaction with the solvent molecules and they move further up the paper than those with a greater affinity for the stationary phase. Paper chromatography is often used to separate the components of an ink, as shown in **Figure 5**. The yellow component of the ink has a greater affinity for the mobile phase and has travelled further up the paper. The purple complement of the ink has a greater affinity for the stationary phase and has travelled the least distance up the paper.

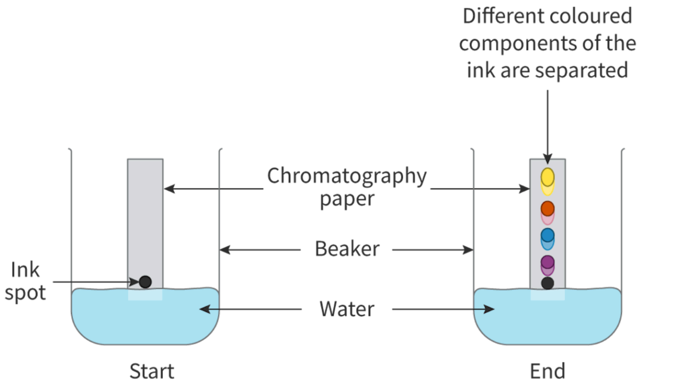


Figure 5. The separation of an ink using paper chromatography.

More information for figure 5

The diagram illustrates a beaker at two stages, labeled 'start' and 'end', demonstrating the process of paper chromatography. At the 'start' stage, a beaker contains a small amount of water, and a strip of chromatography paper is inserted with one end dipped in the water. On the paper, near the surface of the water, is a black ink spot. At the 'end' stage, the black ink spot has separated into four different colored dots, aligned vertically on the paper in the order from top to bottom: yellow, red, blue, and purple, showing the separation of different colored components of the ink. Text annotations label the paper as "Chromatography paper", the liquid as "Water", and the initial mark as "Ink spot". The entire process is described as "Different coloured components of the ink are separated."

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Recrystallisation

Recrystallisation is a separation technique used to remove impurities that are mixed in with a solid. This technique uses the varied solubilities of solids at different temperatures to separate them. The basic principles of the recrystallisation process are as follows. The impure mixture is first dissolved in a small volume of hot solvent which should be just enough to completely dissolve it. At this stage, any insoluble impurities can be filtered off. The solution is then cooled which causes the solubility of the dissolved solids to decrease. The desired product forms crystals leaving the soluble impurities in the solution which is then filtered to obtain the pure product. In this way, the purification of the solid is carried out (**Figure 6**). Recrystallisation is used to purify sugar crystals from sugar cane juice and is also used by pharmaceutical companies to remove any impurities that could contaminate medication.

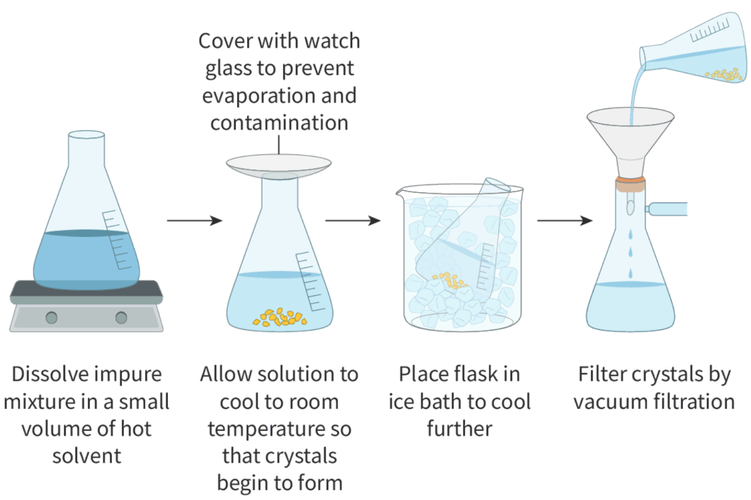

Figure 6. The process of recrystallisation is used to separate a mixture of solids.

More information for figure 6

This diagram illustrates the recrystallisation process through a series of steps: 1. A beaker is placed on a heating device with liquid inside, and the text states, "Dissolve impure mixture in a small volume of hot solvent." 2. An Erlenmeyer flask with a watch glass on top is shown with the instruction, "Cover with watch glass to prevent evaporation and contamination," and the text, "Allow solution to cool to room temperature so that crystals begin to form." Crystals are shown settling at the bottom. 3. The flask is then shown placed in a larger beaker filled with ice, labeled "Place flask in ice bath to cool further." 4. Finally, the diagram shows the solution being poured into a funnel leading to another flask, with the text "Filter crystals by vacuum filtration shown." The diagram illustrates how recrystallisation separates solids from impurities.

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Use the following activity to check your understanding of separation techniques.

Activity

- **IB learner profile attribute**: Knowledgeable
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Part 1

Read each of the following statements and decide if it is correct or not. If it is incorrect, explain to a partner why you think so.

1. If a solution of dirty water is filtered, the filtered water is safe to drink.
2. Clean drinking water can be obtained from seawater by distillation.
3. Filtration can be used to separate a mixture of a solid and a liquid only if the solid is not dissolved in the liquid.
4. Evaporation can be used to separate a mixture of sugar and water.
5. Paper chromatography can be used to separate a mixture of insoluble solids.
6. Recrystallisation can be used to separate a mixture of solids that have the same solubilities at the same temperature.

Part 2

For each of the following mixtures, describe how you would separate them into their individual components. Give a reason for your choice in each case.

1. A solution of aqueous copper sulfate.
2. A solid mixture of insoluble calcium carbonate particles and soluble potassium sulfate particles.
3. A mixture of coloured dyes dissolved in a solution.
4. A mixture of water and ethanol.
5. A mixture of sand and iron filings.

Part 1

1. **Incorrect** — the water could contain dissolved solutes which are not filtered out in the filtration process.
2. **Correct** — seawater contains salt dissolved in water so you could use distillation to separate the salt and water.

3. **Correct** — for filtration to separate a mixture, the solute must not be dissolved in the solvent.
4. **Correct** — the sugar is dissolved in the water so evaporation can be used.
5. **Incorrect** — for paper chromatography to be effective the solutes must be dissolved in the solvent.
6. **Incorrect** — because the solids should have different solubilities at different temperatures in order to separate them using recrystallisation.

Part 2

1. Evaporation because the copper sulfate is dissolved in the water.
2. Solvation of the potassium sulfate in water until it has completely dissolved. Filtration will separate the insoluble calcium carbonate. Evaporate the filtrate to crystallise the potassium sulfate.
3. Paper chromatography because the dyes are dissolved in the solution.
4. Distillation because the two components have different boiling points.
5. A magnet because the iron filings are magnetic but the sand is not.